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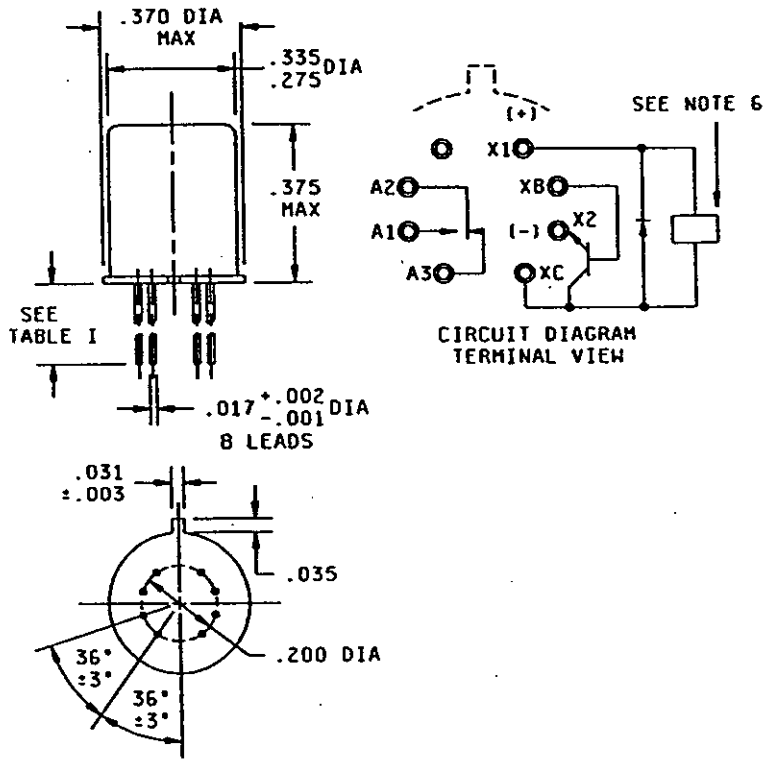
MIL-PRF-28776/4E
10 February 1984
SUPERSEDING
MIL-R-28776/4D
27 September 1982

PERFORMANCE SPECIFICATION SHEET

RELAYS, HYBRID, ESTABLISHED RELIABILITY, SPDT,
LOW LEVEL TO 1.0 AMPERE (SENSITIVE, COIL OPERATE POWER AT 25°C)
(TRANSISTOR DRIVEN) (ELECTROMECHANICAL OUTPUT) WITH DIODE COIL SUPPRESSION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-28776.



Inches	mm	Inches	mm	Inches	mm
.001	0.03	.031	0.79	.335	8.51
.002	0.05	.035	0.89	.370	9.40
.003	0.08	.200	5.08	.375	9.52
.017	0.43	.275	6.98		

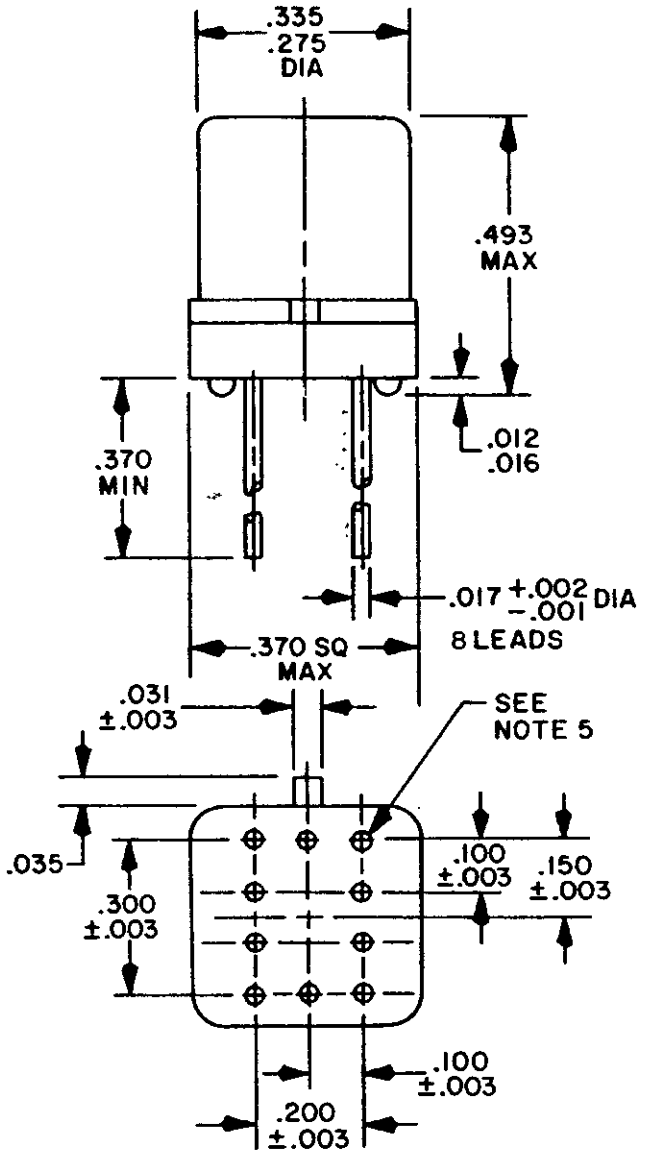
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above are for reference only. Numbers do not appear on relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. Coil symbol optional in accordance with MIL-STD-1285.
7. Circuit diagram shown on part is the terminal view.

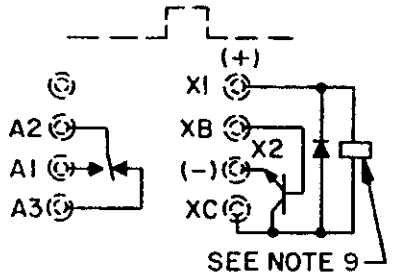
FIGURE 1. Dimensions and configuration.

(E) denotes changes

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CIRCUIT DIAGRAM
TERMINAL VIEW

INCHES	MM
.001	0.03
.002	0.05
.003	0.08
.012	0.30
.016	0.41
.017	0.43
.031	0.79
.035	0.89
.100	2.54
.150	3.81
.200	5.08
.275	6.99
.300	7.62
.335	8.51
.370	9.40
.493	12.52

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Spreader pads shall be certified to MIL-M-38527/5-03.
5. Dimensions and tolerances shown for the bottom view of the spreader pad are for the center to center locations of the holes in the spreader pad.
6. Shape optional within the envelope dimension.
7. Terminal numbers shown above for reference only. Numbers do not appear on relay.
8. Relays shall have a (+) sign placed on circuit diagram as shown.
9. Coil symbol optional in accordance with MIL-STD-1285.
10. Circuit diagram shown on part is the terminal view.

Ⓔ FIGURE 2. Dimensions and configuration relay with spreader pad attached.

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TABLE I. Dash numbers and characteristics. 1/

Dash numbers 2/				Supply voltage (V dc) 5/		At 25°C					Over temperature range			
Lead length			Spreader pad (Fig 2) 4/			Coil resistance (Ohms) ±10%	Pickup voltage (V dc) max	Hold voltage (V dc)	Dropout voltage (V dc)	Max base current to assure turn on 6/ (mA)	Pickup voltage (V dc) max	Hold voltage (V dc)	Dropout voltage (V dc)	Max base current to assure turn on 6/ (mA)
1.500 min 3/	.187 +.040, -.010	.500 min		Rated	Max									
017	025	033	041	5.0	8.0	125	2.6	1.4	0.22	.28	3.6	2.0	0.15	1.20
018	026	034	042	6.0	11.0	255	3.5	2.0	0.28	.20	4.8	2.8	0.18	0.78
019	027	035	043	9.0	16.0	630	5.4	3.0	0.54	.13	7.8	4.2	0.35	0.48
020	028	036	044	12.0	22.0	1,025	6.6	4.0	0.63	.10	10.0	5.6	0.41	0.39
021	029	037	045	18.0	33.0	2,300	9.8	6.0	0.91	.07	14.5	8.4	0.58	0.26
022	030	038	046	26.5	45.0	4,000	12.8	8.0	1.37	.05	19.0	10.4	0.89	0.20
023	031	039	047	32.0	57.0	6,500	18.7	10.0	1.60	.04	24.0	15.0	1.00	0.16
024	032	040	048	40.0	75.0	11,000	23.3	13.0	2.10	.03	30.0	18.7	1.30	0.13

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuit are not recommended for subsequent use in low level applications.

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01.
Example 017L - - - - - 032R.

3/ 1.500 leads are inactive for new design.

4/ Relays supplied with spreader pads (-041 through -048) shall have the pad rigidly attached.

5/ CAUTION: The use of any supply voltage less than the rated supply voltage will compromise the operation of the relay.

6/ Limit base current to 15 mA maximum.

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REQUIREMENTS:

INPUT DATA:

Supply voltage: See table I.

Rated turn-off voltage (or current): 0.3 V dc maximum.

Base turn-on current: See table I.

Control signal voltage (or current): 15 milliamperes maximum.

Coil transient suppression: 1.0 volt maximum.

Turn-off time: 7.5 milliseconds maximum.

Turn-on time: 3.5 milliseconds maximum.

Voltage transients: Applicable.

Diode peak inverse voltage: 100 volts minimum.

CONTACT DATA:

Load ratings:

High level (relay case grounded):

(E) Resistive:
1.0 ampere at 28 V dc.
500 milliamperes at 115 V ac 400 Hz case not grounded.
250 milliamperes at 115 V ac 60 Hz case not grounded.
100 milliamperes at 115 V ac 60 and 400 Hz case grounded.
Inductive: 0.2 ampere at 28 V dc with 0.32 henry inductance.
(E) Lamp: 0.10 ampere at 28 V dc.

Low level: 10 to 50 microamperes at 10 to 50 millivolts dc or peak ac maximum.

Intermediate current: Applicable.

Contact resistance or voltage drop:

(E) Initial: 0.10 ohm maximum, 0.125 ohm maximum with spreader pad attached, (tested at 10 milliamperes maximum at 6 V dc maximum or peak ac).

High level:

During life: Not more than 5 percent of open circuit voltage.
(E) After life: 0.20 ohm maximum (0.225 ohm maximum with spreader pad attached).
During intermediate current: 1.0 ohm maximum.
(E) After intermediate current: 0.20 ohm maximum (0.225 ohm maximum with spreader pad attached).

Low level:

During life: 33 ohms maximum.
(E) After life: 0.15 ohm maximum (0.175 ohm maximum with spreader pad attached).

Contact bounce: 1.5 milliseconds maximum (applicable to failure rate level "L").

Contact stabilization time: 2.0 milliseconds maximum (applicable to failure rate levels "M", "P", and "R").

Overload (high level only): Two times rated current.

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ELECTRICAL DATA:

Insulation resistance: 1/ 10,000 megohms at 500 V dc except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage: 1/

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts both in the energized and deenergized positions	500	125 - All terminals to ca
Between case, frame, or enclosure and coil	500	
Between all contacts and coil	500	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	500	

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Thermal shock: Applicable.

(E) Shock (specified pulse): MIL-STD-202, method 213, test condition B(75 G). Contact chatter shall not exceed 10 microseconds maximum (monitoring of open contact not applicable).

(E) Vibration (sinusoidal): MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum (monitoring of open contact not applicable).

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): Method 1041, MIL-STD-750.

PHYSICAL DATA:

Terminal strength: Method 211, MIL-STD-202:

Pull test: Test condition A, 1-pound pull.
Bend test: Test condition C, 1/2-pound load.
Twist test: Test condition D.

Terminal solderability: Applicable.

Dimensions and configuration: See figures 1 and 2.

(E) Weight: 3.98 grams (0.14 ounce) maximum, 4.23 grams (0.149 ounce) maximum wi spreader attached.

Seal: Hermetic.

1/ Connect input leads together to avoid damage to the relay.

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Internal moisture: Applicable to failure rate levels M, P, and R.

Minimum marking: Military part number, "J" brand, date code, circuit diagram, and manufacturer's name or source code.

LIFE TEST REQUIREMENTS:

(E) High level: 100,000 cycles per relay.
(E) Low level: 100,000 cycles plus 900,000 mechanical life.

PART NUMBER: M28776/4-(dash number from table I and suffix letter designating failure rate level).

QUALIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

TABLE II. Qualification inspection and sample size. 1/

Single submission	Group submission	
18 units plus 1 unsealed unit for level L at C = 0 2/ 33 units plus 1 unsealed unit for level M at C = 0 2/ Qualification inspection as applicable.	M28776/4-038	18 units plus 1 unsealed unit for level L at C = 0 2/ 33 units plus 1 unsealed unit for level M at C = 0 2/ Qualification inspection as applicable.
	M28776/4-033 M28776/4-034 M28776/4-035 M28776/4-036 M28776/4-037 M28776/4-039 M28776/4-040	2 units each part number qualification inspection, group I.

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-28776/3 may be used in addition to MIL-R-28776/4 data.
2/ The number of units required for qualification testing will be increased as required in group IV, table II, MIL-R-28776, if the contractor elects to test the number of units permitting one or more failures.

(E) Initial qualification or relays supplied with spreader pads (-041 through -048) shall be tested as specified below:

Perform the following tests as specified in the qualification inspection table of MIL-R-39016, in the order shown below:

Before installation of pad: screening, visual and mechanical examination (internal), thermal shock, resistance to solvents, shock (specified pulse), acceleration, terminal strength, magnetic interference (when specified), capacitance (when specified), coil life (applicable to continuous duty relays only), resistance to soldering heat, salt spray (corrosion), overload (applicable to high level relays only), life, terminal strength, and intermediate current.

After installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016, in the order shown below:

Insulation resistance, dielectric withstanding voltage, static contact resistance, pickup voltage, hold voltage, dropout voltage, coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

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(E) Qualification inspection (reduced testing for previously qualified relays) for relays supplied with spreader pads (-041 through -048), two (2) units of the 26.5 volt rated coil voltage (-046) shall be tested as follows:

Before installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016 in the order shown below:

For failure rate level "L" only. Screening.
For failure rate levels M, P, and R: Vibration (sinusoidal) test duration shall be 10 minutes, particle impact noise detection (P.I.N.D., when specified), screening.

After installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016 in the order shown below:

Insulation resistance, dielectric withstanding voltage, static contact resistance, pickup voltage, hold voltage, dropout voltage, coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

(E) Group A testing for relays supplied with spreader pads (-041 through -048) shall be tested as specified below:

Before installation of pad perform subgroup 1 of group A tests.
After installation of pad perform subgroups 2 and 3 of group A tests.

Qualification inspection (reduced testing) and sample size: See table III. If the relays produced for MIL-R-28776/4 are similar in construction and design except for the header and coils to the relays produced for MIL-R-28776/3, then reduced testing for qualification of MIL-R-28776/4 relays may be performed concurrent with or subsequent to successful qualification of MIL-R-28776/3 relays.

TABLE III. Qualification inspection (reduced testing).

Examination or test
2 units each supply voltage - Group I of qualification inspection table.
1 unsealed sample unit - Internal examination

SUPERSESSION DATA:

Supersession data: See table IV.

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TABLE IV. Supersession data. 1/

Superseded part number M28776/4-	New part number M28776/4-
001	017
002	018
003	019
004	020
005	021
006	022
007	023
008	024
009	025
010	026
011	027
012	028
013	029
014	030
015	031
016	032

1/ Dash numbers -017 through -024 are inactive for new design and are for support of existing equipment designs only.

Cross reference for Government logistical support: See table V.

TABLE V. Cross reference for Government logistical support.

Superseded part number M28776/4-	New part number M28776/4-	Support with part number M28776/4-	Superseded part number M28776/4-	New part number M28776/4-	Support with part number M28776/4-	New part number M28776/4-	Support with part number M28776/4-
001	017	017	013	029	037	041	041
002	018	018	014	030	038	042	042
003	019	019	015	031	039	043	043
004	020	020	016	032	040	044	044
005	021	021	---	033	033	045	045
006	022	022	---	034	034	046	046
007	023	023	---	035	035	047	047
008	024	024	---	036	036	048	048
009	025	033	---	037	037		
010	026	034	---	038	038		
011	027	035	---	039	039		
012	028	036	---	040	040		

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Air Force - 85

Review activities:
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DLA - ES
NSA - NS

User activities:
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Air Force - 19

Agent:
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Preparing activity:
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